

CSE314 Bash Scripting Online - Section C

May 12, 2025

Website Deployment Packager

Time: 40 minutes

Overview

As a junior web developer at TechFront Solutions, you've been assigned to the final phase of a client project. The client's website is complete and tested, but now needs to be prepared for deployment to their production server. The current development structure in the **website/** directory is disorganized, with files scattered across various subdirectories. Your manager has tasked you with writing a bash script that will reorganize these files into an optimized deployment structure while maintaining all functionality.

This script will become part of the company's deployment pipeline, ensuring consistent website deployments for all future clients. Your ability to complete this task efficiently and accurately will demonstrate your value to the team and potentially lead to more responsibilities in future projects.

Objectives

1. Scan the **website/** directory recursively for all web files.
2. Categorize files by type into:
 - HTML files (**.html**, **.htm**)
 - Stylesheets (**.css**)
 - Scripts (**.js**)
 - Images (**.jpg**, **.png**, **.gif**, **.svg**)
3. Create a **deploy/** directory with proper structure:
 - HTML files in root (the **deploy/** directory)
 - CSS in **assets/css/**
 - JS in **assets/js/**
 - Images in **assets/images/**
4. Update all internal links in HTML files to match the new structure.

Sample Directory Structure

📁 website/ (Input directory)

📁 styles/

- 📄 main.css
- 📄 about.css

📁 scripts/

- 📄 main.js
- 📄 contact.js

📁 images/

- 📄 logo.png
- 📄 banner.jpg

📁 pages/

- 📄 about.html
- 📄 contact.html

📄 index.html

📁 deploy/ (Output directory)

📄 index.html

📄 about.html

📄 contact.html

📁 assets/

📁 css/

- 📄 main.css
- 📄 about.css

📁 js/

- 📄 main.js
- 📄 contact.js

📁 images/

- 📄 logo.png
- 📄 banner.jpg

Command

```
./website_packager.sh website/
```

Requirements

- Your script must scan the `website/` directory recursively.
- HTML files should be copied to the root of the `deploy/` directory, regardless of their original location.
- All CSS files should be copied to `deploy/assets/css/`.
- All JavaScript files should be copied to `deploy/assets/js/`.
- All image files should be copied to `deploy/assets/images/`.
- All links within HTML files must be updated to point to the new file locations.
- If the `deploy/` directory already exists, it should be recreated before starting.
- Your script should handle files with spaces and special characters in their names.

File Modification Details

You need to update the following HTML elements based on these simple rules:

- **CSS files:** Change any path to CSS files to `assets/css/filename.css`
Example: `styles/main.css` or `../styles/main.css` becomes `assets/css/main.css`
- **JavaScript files:** Change any path to JS files to `assets/js/filename.js`
Example: `scripts/main.js` or `../scripts/contact.js` becomes `assets/js/main.js`
- **Image files:** Change any path to image files to `assets/images/filename.ext`
Example: `images/logo.png` or `../images/banner.jpg` becomes `assets/images/logo.png`
- **HTML files:** All HTML files should be referenced directly at the root
Example: `pages/about.html` or `../about.html` becomes `about.html`

Hints

- Use the `find` command to locate files by extension.
- Consider using `sed` for updating file paths in HTML files.
- Test your script with both absolute and relative paths.
- For path replacements in HTML files, focus on the file extension to determine the appropriate folder:
 - `.css` files → `assets/css/`
 - `.js` files → `assets/js/`
 - `.jpg`, `.png`, `.gif`, `.svg` files → `assets/images/`
 - `.html` files → root directory
- Simple `sed` commands with basic pattern matching will be sufficient for this task.

Submission Guidelines

Important Notes

- Please note that any usage of the internet is strictly prohibited during the assignment.
- Use of any unfair means will be duly punished.
- Submit the source code in Moodle. Failure to submit in Moodle will result in a mark deduction.
- The filename should be `online-StudentID.sh` (e.g., `online-2105XXX.sh`).
- Make sure your script is executable.